Manufacturing Report
Group 10

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Initial Design

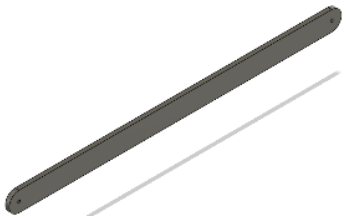


Figure 1 - Initial Design

Figure 1 represents the initial design for a linkage bar in the linkage array, where it would be produced with the array's functionality in mind. This design was also informed by the requirements and targets set out in PDS items: 11, 25, 21. Design for Manufacture was the most prominent of these targets and largely governed the initial design. Resulting design constraints included: having a constant cross section thereby allowing for 2D profile cutting processes such as laser cutting, waterjet cutting and CNC milling. Having no internal cavities allowed for methods like turning, die casting or injection moulding and simplifying geometry where possible. Aesthetic considerations were low influence due to the functional nature of the part. Design was also influenced by considerations concerning reducing weight and cost.

Process Selection

Manufacturing Process PRIMA Selection matrix (1) was used to first identify the most suitable manufacturing processes for our chosen material of stainless steel and our estimated production scale of 1,000 – 10,000 units made (relevant PDS items: 26, 11). From an initial large spectrum of manufacturing processes, processes such as closed die forging and powder metallurgy can be immediately discarded due to being too cost inefficient. Due to production quantity, methods such as electrochemical machining and electron beam machining would impose large equipment and tooling costs considering production scale. Processes that limit the part's structural integrity can also be discarded such as powder metallurgy which causes steel to become more porous and weak. Some forms of machining can be discarded due to the thickness limitations of processes such as laser beam machining and electron beam machining. The remaining processes: **Automatic Machining**, **Ceramic Mould Casting** and **Shell Moulding** were then comparatively evaluated against the following categories in Table 1: **Cost**, **Lead time**, **Quality Control**, **Dimensional Accuracy**, **Design Flexibility**. (2)(3)(4)

Table 1 - Comparison between Manufacturing Processes

Manufacturing Process	Cost		Lead Time		Design Flexibility	Dimensional Accuracy		Total Score
	Cost per Unit	Score	Time Frame	Score	Score	Tolerances in mm	Score	
Automatic Machining	£7.12	0	Short	0	0	±0.051	0	0
Ceramic Mould Casting	£7.45	-1	Short	0	-1	±0.25	-1	-3
Shell Moulding	£7.45	-1	Moderate	-1	-1	±0.25	-1	-4

Cost: per unit cost was estimated using equation [1]:

$$c_u = \left(R_p \cdot t_u\right) \cdot \frac{1}{3600} + c_m \quad [1]$$

Where c_u represents per unit cost in £, R_p represents processing rate in £/hour, t_u represents unit time in seconds and c_m represents material cost in £. Processing rate was found according to equation [2]:

$$R_p = c_l + r_m \quad [2]$$

Where c_l represents labour cost rate and r_m represents machine cost rate, both in £/hour. Material costs were calculated using equation [3]:

$$c_m = M \cdot c_s \quad [3]$$

Where Mrepresents stock mass in kg and $c_s$represents material cost in $£/kg$.

A more accurate model may have included factors such as failure rate, byproduct disposal or other overhead costs however these were deemed negligible due to being consistent across processes.

Lead Time: Lead times vary significantly depending on the size and form of the part, however for comparison, Lead time estimates provided in the manufacturing process selection handbook (1) were used.

Design Flexibility: A qualitative comparison considering the initial design was carried out, finding both casting-based methods (Shell Moulding and Ceramic Mould casting) to be equivalent with Automatic Machining being deemed most suitable due to the fixed cross section of the part and its simple geometry.

Dimensional Accuracy: Dimensional accuracies are quoted from the manufacturing process selection handbook (1). In CNC milling, dimensional accuracies can vary therefore tolerances have been estimated using the milling process capability chart (1).

From this Automatic machining (CNC Milling) was selected and design iterations were carried out on the part to improve suitability for CNC Milling.

Design Iteration

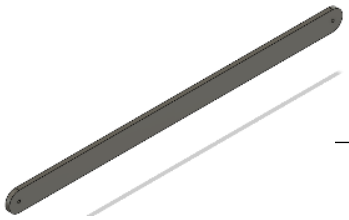


Figure 2 - Initial Design

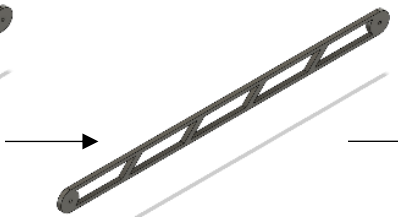


Figure 3 – Iterated Design 1

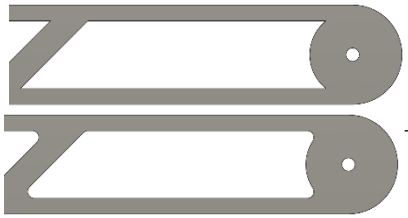


Figure 4 – Iterated Design 2



Figure 5 – Iterated Design 3

The initial design shown in figure 2 adheres fully to all the constraining elements of the design, however additional considerations such as minimizing mass and cost allow for more suitable iterations (relevant PDS items: 21,23,25,26,11,13,15,9) .The first of these iterations (figure 3) employs cutout optimization whereby smaller sections of the part are removed in a manner that preserves structural integrity. Here structural integrity is preserved by a series of five 45° struts as well as additional reinforcement, radially surrounding the holes around which the part will rotate. Due to the use of 15mm stock 316 stainless steel, the part will remain functional despite the material loss which corresponds to a mass reduction of 31.2%.

This iteration however introduces the issue of points of very high stress especially under loading as well as reducing suitability for CNC milling due to a lack of rounded edges, which are necessary due to the kerf of any milling bit used in removing the cutouts.

The 5mm fillet shown in figure 4, greatly mitigate this concern as well as allowing for milling with a bit as large as 5mm thereby significantly increasing ease of manufacture.

For post processing on milled stainless steel, it is necessary to remove material from the part where sharp edges (along the milled edges of the part) therefore deburring will be necessary as well as a 10mm fillet on the outer edges that are most likely to be interacted with during regular use.

The softer edges of the part shown in figure 5 greatly increase operational safety especially considering the 1200mm swept radius of the part during the opening and closing of the mechanism. This will decrease the ease of manufacture in requiring the additional use of a 10mm round over bit.

Post Processing and Surface Finish

Once milled, the produced part will be cleaned using an alkaline cleaning solution to remove residual oils, dirt, coolant and any other possible contaminants as well as adequately prepare the part surface for finishing. Following this, the part will be deburred, and quality assurance will be carried out to ensure all measured part dimensions fall within the provided tolerances.

In selecting surface finish and post processing methods, corrosion resistance was identified as the most influential in extending part lifespan and performance. (relevant PDS items: 25,12,26). with the corrosion class C3 selected as a normal use case, where methods such as electropolishing paint, protection systems and galvanization were selected.

Paint Protection: Lowest price per square meter ($31.40) (5) assuming an epoxy finish, however despite being the cheapest option Paint Protection Systems are the most susceptible to damage in the form of chips due to impact and general wear due to abrasion. It also ranks lowest in durability requiring replacement every 5 to 10 years in **ideal** conditions. (7)

Galvanization: Highest price per square meter ($56.85) (5) requiring replacement once every 15 to 20 years under **normal** conditions with the highest wear resistance of the selected methods. (6)

Electropolishing: It is difficult to source a price per square meter as this can vary largely depending on the part, however due to the relatively simple geometry of the part and the use of stainless-steel, price should fall somewhere in between the cheapest and most expensive considered options. It is also particularly resistant to moisture due to removing surface imperfections that would otherwise act as corrosion sites thereby ranking first in terms of corrosion resistance. (7)

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